

Cheat-sheet — keep this at the stand (English)

One page. Print it. German: [DE_spickzettel.md](#). Full detail: [EN_games-guide.md](#), [EN_quantum-basics.md](#).

The 3 big ideas (the whole stand in 3 lines)

1. **Superposition** — a quantum thing can be in two states at once (spinning coin).
2. **Measurement** — looking forces it to pick one, at **random**; you can't predict which.
3. **Quantum randomness is a resource** — truly unpredictable numbers power security, AI, new computers.

30-second pitches

- **Hack the Light:** "Turn light into a 0/1. First you control it, then a hidden rule, then pure noise you can't steer — that's how we harvest randomness."
- **Beat the AI:** "Tap left/right; the AI predicts you and wins. Humans can't be random — that's why we need quantum randomness."
- **Quantum Penalty:** "Beat a mind-reading keeper. The quantum shot is in two corners at once and only 'decides' when it arrives — he can't read what isn't decided yet."
- **Quantum Chess:** "Chess where a piece can be in two squares at once (superposition), get entangled, and 'collapse' to one square when challenged."
- **Quantum Tic-Tac-Toe:** "Your mark goes in two squares at once; tangled marks collapse into real ones. Superposition + entanglement + measurement in 30 seconds."

Age one-liners (the same arc)

- **6-8:** "First you control the machine. Then it gets sneaky. Then nature wins."
- **9-12:** "Some rules are simple, some hide information, some things can't be predicted at all."
- **13+:** "AI finds patterns; where there's none, prediction fails — that's where quantum randomness lives."

Honesty (if asked "is this *really* quantum?")

- Be proud and honest: **these are quantum-inspired games and a physical random source** (camera noise), not quantum hardware. They build the *right intuition*; the real tech uses single photons.
- Never claim there's a quantum computer or single-photon device on the stand.

Top FAQ

- "What is quantum, simply?" → "The rules for the tiniest things: they can be in two states at once, and looking forces a random choice nobody can predict."
- "Why does randomness matter?" → "Every password and secure chat needs unpredictable keys. Predictable = hackable. Quantum gives the most unpredictable numbers there are."
- "Can entanglement send messages faster than light?" → "No — it's a correlation, not a phone."
- "Is a quantum computer just faster?" → "Different, and only faster for certain problems."
- "How do I beat the AI / quantum keeper?" → "Be truly random — which humans basically can't be."

Practical

- Each turn 30-90 s. Queue plays on their **own phones** via the **QR** (arraxis.com/nachtdertechnik).
- **Leaderboard** keeps each player's best score + attempts (boards: penalty, beat-ai, light).
- **Quanten-Geschenk:** top players → open [prize.html](#), press the button, hand out the sweet.
- Camera game needs permission; if denied it has a no-camera demo. Keep a **tablet** ready for it.

- Hygiene: hand sanitiser out; wipe shared screens now and then.
- If anything web breaks: external games (q-chess.com, tiqtaqtoe.com) and the camera game's demo mode are independent fallbacks.